

Adinkra Microfinance Ltd - Digital Forensics Report

Case/Reference No.: HIVE-2026-BF-0117

Analyst: Julian

Date: 20 July 2026

Classification: Training / portfolio exercise. synthetic data (HIVE CONSULT exhibit, not a real incident)

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1. Executive Summary

Adinkra Microfinance Ltd (Accra) suspected that a staff accountant, Kwabena Asare, who worked on workstation FIN-WS-07, had been job-hunting on company time and had moved a confidential client list out of the organization ahead of resigning to start a competing practice. A USB flash drive linked to the subject was seized and imaged for examination. Analysis of that image found browser, document, and receipt evidence consistent with the allegation: the subject searched for new jobs and for how to export company data, uploaded a client spreadsheet and a QuickBooks backup to a personal Dropbox account, emailed himself a backup link, registered a competing business, purchased a personal laptop, and drafted a resignation letter, all within a five-day window (15 to 20 September 2023). No evidence of anti-forensic activity, such as timestamp manipulation, was found. An apparent date/time inconsistency identified during initial review was traced to a display limitation in the examination tool rather than to the underlying evidence, and is documented in Finding 5.10.

2. Scope & Objectives

- **Scope:** one USB flash drive image (HIVE-2026-BF-0117-USB.E01), FAT32 volume KASARE_USB, containing a preserved Google Chrome profile and personal documents, receipts, and images belonging to the subject.
- **Objectives:** reconstruct the subject's web browsing activity; determine whether that activity is corroborated by on-disk documents and receipts; establish whether data was exfiltrated and by what channel; build a timeline of the subject's actions between 15 and 20 September 2023.
- **Out of scope:** no \$MFT, \$LOGFILE, \$USNJRNL, or Windows registry hives were available, the volume is FAT32, not NTFS, so these artifacts do not exist on this drive. No live system or memory analysis was performed, and no network-level evidence was examined.

3. Evidence & Chain of Custody

Item #	Description	Source	Hash (SHA-256)	Date Acquired	Acquired By
1	HIVE-2026-BF-0117-USB.E01 (E01 disk image)	Synthetic training exhibit, HIVE CONSULT	Container: 00B16EF7B2E3707C09C0F97B56285F653A33011545A45813450BC6F89A131A07 Media MD5: f1336b78c7b9771d0f1c6c78c116b479 Media SHA1: 6f3aba0cb15869372dd1311674e9a353e7ec641a	14 Jul 2026, 10:11:18 (per E01 metadata)	HIVE CONSULT (synthetic acquisition)

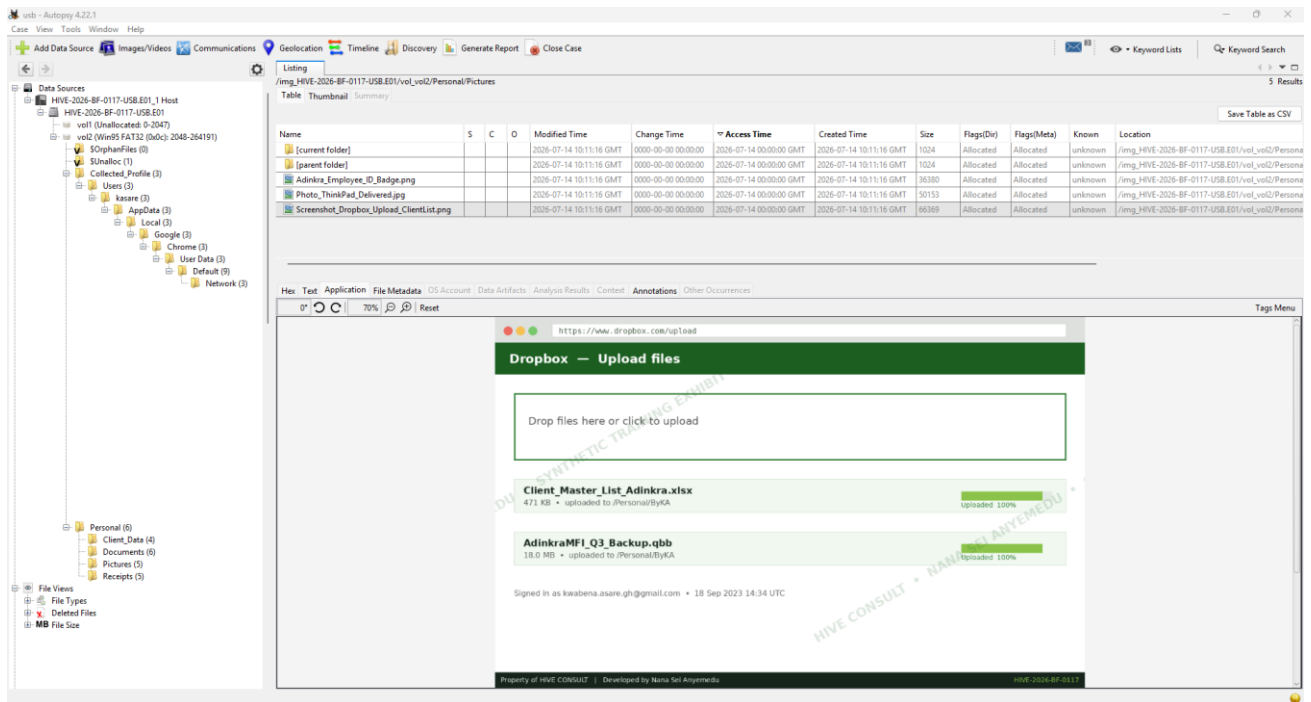
Note: this is a synthetic training exhibit, not real custody evidence. The hash values above were independently recalculated against the supplied image and match the values recorded at acquisition, confirming the file examined is unaltered.

```
Administrator: Windows PowerShell
PS C:\Users\CyberSamurai\Desktop\DFIR Cases\USB Forensics\HIVE-2026-BF-0117-USB_lab>
PS C:\Users\CyberSamurai\Desktop\DFIR Cases\USB Forensics\HIVE-2026-BF-0117-USB_lab> Get-FileHash "C:\Users\CyberSamurai\Desktop\DFIR Cases\USB Forensics\HIVE-2026-BF-0117-USB.E01" -Algorithm SHA256

Algorithm Hash Path
-----
SHA256 00B16EF7B2E3707C09C0F97B56285F653A33011545A45813450BC6F89A131A07 C:\Users\CyberSamurai\Desktop\DFIR Cases\USB Forensics\HIVE-2026-B...

PS C:\Users\CyberSamurai\Desktop\DFIR Cases\USB Forensics\HIVE-2026-BF-0117-USB_lab>
```

Independent hash verification of the container file.



USB drive content listing (Personal folder: Client_Data, Documents, Pictures, Receipts).

4. Tools & Methodology

Tool	Version	Purpose
Autopsy	4.22.1	Primary GUI triage: Web History, File Metadata, Data Artifacts
Sleuth Kit / libewf	current (apt)	Direct image mounting and verification (mmls, fsstat, fls, icat)
sqlite3	3.45.1	Direct query of the Chrome History database
Python 3	3.x	WebKit-epoch timestamp decoding and cross-checks

Methodology:

- Verify container integrity against the published MD5/SHA1 (media) and SHA256 (container) hashes before analysis.
- Mount the E01 (ewfmount) and confirm the partition and filesystem layout (mmls, fsstat).
- Triage the Chrome profile in Autopsy (Web History, Login Data, Downloads) to reconstruct browsing activity.
- Where Autopsy's raw SQLite table view produced anomalous timestamps, extract the History database directly (icat) and re-derive timestamps using the WebKit epoch formula, cross-checked against a purpose-written SQL query.
- Correlate browser artifacts against on-disk documents, receipts, and screenshots recovered from the same image.
- Build a consolidated timeline and validate each finding against at least one independent artifact.

5. Findings

5.1 Last known activity

Observation: The last page the subject visited before the browsing session ended was the Formula 1 “Latest News” page.

Evidence: Web History: title “Latest News”, URL <https://www.formula1.com/en/latest.html>, timestamp 2023-09-19 09:58:08 UTC. Source path: /Users/kasare/AppData/Local/Google/Chrome/User Data/Default/History.

The screenshot displays the Autopsy 4.22.1 interface. On the left, the 'Data Sources' pane shows a tree structure with 'HIVE-2026-BF-0117-USB.E01_1 Host' expanded, revealing 'vol1 (Unallocated: 0-2047)' and 'vol2 (WinFS FAT32 (Indoc): 2048-264191)'. The 'vol2' drive contains folders like 'SOrphanFiles (0)', 'SUnalloc (1)', 'Collected_Profile (3)', 'Users (3)', 'AppData (3)', 'Local (3)', 'Google (3)', 'Chrome (3)', 'User Data (3)', 'Default (9)', 'Network (3)', 'Personal (6)', 'Client_Data (4)', 'Documents (6)', 'Pictures (5)', and 'Receipts (3)'. The 'File Views' pane on the right shows 'File Types', 'Deleted Files', 'MB File Size', 'Data Artifacts', 'Analysis Results', 'OS Accounts', 'Tags', 'Score', and 'Reports'. The main pane is divided into two sections. The top section, 'Listing', shows a table of file metadata for the 'History' folder. The bottom section, 'Table', shows a list of web history entries. The entry for 'https://www.formula1.com/en/latest.html' is highlighted with a red box.

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Location	MD5 Hash	SHA-256
History				2026-07-14 10:11:16 GMT	0000-00-00 00:00:00	2026-07-14 00:00:00 GMT	2026-07-14 10:11:16 GMT	1024	Allocated	Allocated	unknown	/img_HIVE-2026-BF-0117-USB.E01/vol2/Collected_Profile/Users/kasare/AppData/Local/Google/Chrome/User Data/Default/History		
parent folder				2026-07-14 10:11:16 GMT	0000-00-00 00:00:00	2026-07-14 00:00:00 GMT	2026-07-14 10:11:16 GMT	1024	Allocated	Allocated	unknown	/img_HIVE-2026-BF-0117-USB.E01/vol2/Collected_Profile/Users/kasare/AppData/Local/Google/Chrome/User Data/Default/History		
Bookmarks				2026-07-14 10:11:16 GMT	0000-00-00 00:00:00	2026-07-14 00:00:00 GMT	2026-07-14 10:11:16 GMT	1905	Allocated	Allocated	unknown	/img_HIVE-2026-BF-0117-USB.E01/vol2/Collected_Profile/Users/kasare/AppData/Local/Google/Chrome/User Data/Default/History		
Favorites				2026-07-14 10:11:16 GMT	0000-00-00 00:00:00	2026-07-14 00:00:00 GMT	2026-07-14 10:11:16 GMT	24576	Allocated	Allocated	unknown	/img_HIVE-2026-BF-0117-USB.E01/vol2/Collected_Profile/Users/kasare/AppData/Local/Google/Chrome/User Data/Default/History		
Login Data				2026-07-14 10:11:16 GMT	0000-00-00 00:00:00	2026-07-14 00:00:00 GMT	2026-07-14 10:11:16 GMT	20480	Allocated	Allocated	unknown	/img_HIVE-2026-BF-0117-USB.E01/vol2/Collected_Profile/Users/kasare/AppData/Local/Google/Chrome/User Data/Default/History		
Preferences				2026-07-14 10:11:16 GMT	0000-00-00 00:00:00	2026-07-14 00:00:00 GMT	2026-07-14 10:11:16 GMT	2032	Allocated	Allocated	unknown	/img_HIVE-2026-BF-0117-USB.E01/vol2/Collected_Profile/Users/kasare/AppData/Local/Google/Chrome/User Data/Default/History		
Web Data				2026-07-14 10:11:16 GMT	0000-00-00 00:00:00	2026-07-14 00:00:00 GMT	2026-07-14 10:11:16 GMT	20480	Allocated	Allocated	unknown	/img_HIVE-2026-BF-0117-USB.E01/vol2/Collected_Profile/Users/kasare/AppData/Local/Google/Chrome/User Data/Default/History		

id	url	title	visit_count	typed_count	last_visit_time	hidden
6	https://www.formula1.com/en/latest.html	Latest News	2	0	424684/10/24 06:53:20	0
5	https://www.formula1.com/en/latest.html	F1 - The Official Home of Formula 1® Racing	2	0	424684/10/24 06:40:00	0
23	https://www.google.com/search?q=formula+1&sa=s&vs=56561006&sources=hp	Formula 1 - Google Search	1	0	424684/10/24 04:10:00	0
7	https://www.google.com/	Google	4	4	424684/10/24 01:56:40	0
1	https://www.imdb.com/	IMDb: Ratings, Reviews, and Where to Watch the Best Movies & TV Shows	2	2	424684/10/23 23:26:40	0
8	https://www.netflix.com/gh/	Netflix Ghana - Watch TV Programmes Online, Watch Films Online	3	3	424684/10/23 20:23:20	0
4	https://www.dropbox.com/home	Home - Dropbox	4	1	424684/10/23 04:00:00	0
22	https://www.dropbox.com/	Dropbox	1	1	424684/10/23 04:00:00	0
21	https://outlook.live.com/owa/	Microsoft Outlook Personal Email and Calendar Microsoft 365	1	1	424684/10/22 20:46:40	0
18	https://accounts.google.com/v3/signin/identifier/continue=https%3A%2F%2Fmail.google.com	Gmail	1	1	424684/10/22 19:06:40	0
19	https://accounts.google.com/ServiceLogin?service=mail	Gmail	1	0	424684/10/22 19:06:40	0
20	https://mail.google.com/mail/u/0/	Inbox - kwebena.asare.gh@gmail.com - Gmail	1	0	424684/10/22 16:36:40	0
17	https://www.ebay.com/km/254654754368?hash=item3b4a9b2e40	Lenovo ThinkPad P50s Laptop 15.6" Core i7 16GB Ram 512GB SSD Win10 - E\$89.99	1	0	424684/10/22 14:56:40	0
16	https://www.ebay.com/sch/i.html?_from=R40&_trksid=p4432023.m570.1313&_nkw=laptop+L...	Laptop for sale eBay	1	0	424684/10/22 12:40:00	0
15	https://www.google.com/search?q=starting+an+accounting+firm+in+ghana	starting an accounting firm in ghana - Google Search	1	0	424682/08/07 11:33:20	0
14	https://www.dropbox.com/upload	Upload files - Dropbox	1	0	424682/08/07 11:33:20	0
13	https://www.google.com/search?q=how+to+export+quickbooks+company+data	how to export quickbooks company data - Google Search	1	0	424673/11/28 07:00:00	0
12	https://www.jobberman.com.gh/jobs?q=accountant	Accountant Jobs in Ghana Jobberman	1	0	424673/11/28 07:00:00	0

History file location and metadata.

Interpretation: This is the final entry in the subject's browsing history and marks the end of the reconstructed session on 19 September 2023.

5.2 Exit-plan browsing pattern

Observation: Five search terms in the history indicate the subject was researching new employment and independently starting a competing accounting practice.

Evidence: Web History url/title table:

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Location	MDS Hash	SHA-256
[current folder]				2026-07-14 10:11:16 GMT	0000-00-00 00:00:00	2026-07-14 00:00:00 GMT	2026-07-14 10:11:16 GMT	1024	Allocated	Allocated	unknown	/img_HIVE-2026-BF-0117-USB.E01/vol2/Collected_Profile/Users/kasare/AppData/Local/Google/Chrome/User Data/Default		
[parent folder]				2026-07-14 10:11:16 GMT	0000-00-00 00:00:00	2026-07-14 00:00:00 GMT	2026-07-14 10:11:16 GMT	1024	Allocated	Allocated	unknown	/img_HIVE-2026-BF-0117-USB.E01/vol2/Collected_Profile/Users/kasare/AppData/Local/Google/Chrome/User Data/Default		
History				2026-07-14 10:11:16 GMT	0000-00-00 00:00:00	2026-07-14 00:00:00 GMT	2026-07-14 10:11:16 GMT	1905	Allocated	Allocated	unknown	/img_HIVE-2026-BF-0117-USB.E01/vol2/Collected_Profile/Users/kasare/AppData/Local/Google/Chrome/User Data/Default		
Login Data				2026-07-14 10:11:16 GMT	0000-00-00 00:00:00	2026-07-14 00:00:00 GMT	2026-07-14 10:11:16 GMT	24576	Allocated	Allocated	unknown	/img_HIVE-2026-BF-0117-USB.E01/vol2/Collected_Profile/Users/kasare/AppData/Local/Google/Chrome/User Data/Default		
Preferences				2026-07-14 10:11:16 GMT	0000-00-00 00:00:00	2026-07-14 00:00:00 GMT	2026-07-14 10:11:16 GMT	20480	Allocated	Allocated	unknown	/img_HIVE-2026-BF-0117-USB.E01/vol2/Collected_Profile/Users/kasare/AppData/Local/Google/Chrome/User Data/Default		
Web Data				2026-07-14 10:11:16 GMT	0000-00-00 00:00:00	2026-07-14 00:00:00 GMT	2026-07-14 10:11:16 GMT	20480	Allocated	Allocated	unknown	/img_HIVE-2026-BF-0117-USB.E01/vol2/Collected_Profile/Users/kasare/AppData/Local/Google/Chrome/User Data/Default		

id	url	title	visit_count	typed_count	last_visit_time	hidden
8	https://www.netflix.com/gh/	Netflix Ghana - Watch TV Programmes Online, Watch Films Online	3	3	424684/10/23 20:23:20	0
4	https://www.dropbox.com/home	Home - Dropbox	4	1	424684/10/23 04:00:00	0
22	https://www.dropbox.com/	Dropbox	1	1	424684/10/23 04:00:00	0
21	https://outlook.live.com/owa/	Microsoft Outlook Personal Email and Calendar Microsoft 365	1	1	424684/10/22 20:46:40	0
18	https://accounts.google.com/v3/signin/identifier?continue=https%3A%2F%2Fmail.google.com	Gmail	1	1	424684/10/22 19:06:40	0
19	https://accounts.google.com/ServiceLogin?service=mail	Gmail	1	0	424684/10/22 19:06:40	0
20	https://mail.google.com/mail/u/0/	Inbox - kwabena.asare.gh@gmail.com - Gmail	1	0	424684/10/22 19:06:40	0
17	https://www.ebay.com/itm/254654754386?hash=item3b4a9b2e40	Lenovo ThinkPad P50s Laptop 15.6" Core i7 16GB Ram 512GB SSD Win10 - £599.99 eBay	1	0	424684/10/22 16:36:40	0
16	https://www.ebay.com/sch/1.html?from=R40&trksid=p4432023.m570.1313&_nkw=laptop+f...	laptop for sale eBay	1	0	424684/10/22 14:56:40	0
15	https://www.google.com/search?q=starting+an+accounting+firm+in+ghana	starting an accounting firm in ghana - Google Search	0	0	424680/08/22 12:40:00	0
14	https://www.dropbox.com/upload	Upload files - Dropbox	0	0	424680/08/22 11:33:20	0
13	https://www.google.com/search?q=how+to+export+quickbooks+company+data	how to export quickbooks company data - Google Search	0	0	424680/08/22 11:33:20	0
12	https://www.jobberman.com.gh/jobs?q=accountant	Accountant Jobs in Ghana Jobberman	0	0	424679/11/28 07:06:00	0
11	https://www.linkedin.com/jobs/	Find Jobs in Ghana Jobberman Ghana	1	1	424679/11/25 16:30:00	0
10	https://www.linkedin.com/jobs/	Jobs LinkedIn	0	0	424679/11/16 23:46:40	0
9	https://www.google.com/search?q=forensic+accounting+jobs+accra	forensic accounting jobs accra - Google Search	0	0	424679/11/16 11:40:00	0
3	https://www.dropbox.com/login	Sign in - Dropbox	1	0	424625/07/28 07:23:20	0
2	https://www.google.com/search?q=dropbox+login	dropbox login - Google Search	1	0	424625/07/28 00:26:40	0

Search-term evidence from Web History.

Interpretation: The combination of job-search terms and a business-formation search shows deliberate research into both leaving the company and competing with it, not incidental browsing.

URL	Title
google.com/search=starting+an+accounting+firm+in+ghana	starting an accounting firm in ghana - Google Search
jobberman.com.gh/jobs=accountant	Accountant Jobs in Ghana Jobberman
jobberman.com.gh/	Find Jobs in Ghana Jobberman Ghana
linkedin.com/jobs/	Jobs LinkedIn
google.com/search?q=forensic+accounting+jobs+accra	forensic accounting jobs accra - Google Search

5.3 Personal cloud storage and initial access (Dropbox)

Observation: Dropbox was the personal cloud service used by the subject, first accessed 2023-06-20 09:09:05 UTC and visited eight times in total.

Evidence: Attribution is established two ways: the URL directory pattern (dropbox.com/upload) and the saved Chrome credential for kwabena.asare.gh@gmail.com against dropbox.com/login. The first-access timestamp is independently corroborated by the Dropbox Plus payment receipt, billing date 20 Jun 2023.

Table: urls 23 entries Page 1 of 1 Export to CSV							
id	url	title	visit_count	typed_count	last_visit_time	hidden	
1	https://www.google.com/	Google	4	4	13339991063000000	0	
2	https://www.google.com/search?q=dropbox+login	dropbox login - Google Search	1	0	13331725720000000	0	
3	https://www.dropbox.com/login	Sign in - Dropbox	1	0	13331725745000000	0	
4	https://www.dropbox.com/home	Home - Dropbox	4	1	13339990984000000	0	
5	https://www.formula1.com/	F1 - The Official Home of Formula 1® Racing	3	2	13339991080000000	0	
6	https://www.formula1.com/en/latest.html	Latest News	2	0	13339991080000000	0	
7	https://www.imdb.com/	IMDb: Ratings, Reviews, and Where to Watch the Best Movies & TV Shows	2	2	13339991043000000	0	
8	https://www.netflix.com/gh/	Netflix Ghana - Watch TV Programmes Online, Watch Films Online	3	3	13339991043000000	0	
9	https://www.google.com/search?q=forensic+accounting+jobs+accra	forensic accounting jobs accra - Google Search	1	0	13339245930000000	0	
10	https://www.linkedin.com/jobs/	Jobs LinkedIn	1	0	13339245970000000	0	
11	https://www.jobberman.com.gh/	Find Jobs in Ghana Jobberman Ghana	1	1	13339246725000000	0	
12	https://www.jobberman.com.gh/jobs?q=accountant	Accountant Jobs in Ghana Jobberman	1	0	13339246950000000	0	
13	https://www.google.com/search?q=how+to+export+quickbooks+company+data	how to export quickbooks company data - Google Search	1	0	13339519840000000	0	
14	https://www.dropbox.com/upload	Upload files - Dropbox	1	0	13339521200000000	0	
15	https://www.google.com/search?q=starting+an+accounting+firm+in+ghana	starting an accounting firm in ghana - Google Search	1	0	13339522500000000	0	
16	https://www.ebay.com/sch/i.html?from=R40&trksid=p4432023.m570.11313&_nkw=laptop...	laptop for sale eBay	1	0	13339990937000000	0	
17	https://www.ebay.com/itm/254654754368?hash=item3b4a9b2e40	Lenovo ThinkPad P50s Laptop 15.6" Core i7 16GB Ram 512GB SSD Win10 - £589.99 eBay	1	0	13339990943000000	0	
18	https://accounts.google.com/v3/signin/identifier?continue=https%3A%2F%2Fmail.google...	Gmail	1	1	13339990952000000	0	
19	https://accounts.google.com/ServiceLogin?service=mail	Gmail	1	0	13339990952000000	0	
20	https://mail.google.com/mail/u/0/	Inbox - kwabena.asare.gh@gmail.com - Gmail	1	0	13339990952000000	0	
21	https://outlook.live.com/owa/	Microsoft Outlook Personal Email and Calendar Microsoft 365	1	1	13339990958000000	0	
22	https://www.dropbox.com/	Dropbox	1	0	13339990984000000	0	
23	https://www.google.com/search?q=formula+1&srca_esvs=566561006&source=hp	formula 1 - Google Search	1	0	13339991071000000	0	

Dropbox visit count from Web History.

Interpretation: Dropbox was accessed 8 times as shown by tally of corresponding values in the visit_count fields.

The page that indicates files were move into the service is <https://www.dropbox.com/upload>

8	https://www.netflix.com/gh/	Netflix Ghana - Watch TV Programmes Online, Watch Films Online	3	3	13339991043000000	
9	https://www.google.com/search?q=forensic+accounting+jobs+accra	forensic accounting jobs accra - Google Search	1	0	13339245930000000	
10	https://www.linkedin.com/jobs/	Jobs LinkedIn	1	0	13339245970000000	
11	https://www.jobberman.com.gh/	Find Jobs in Ghana Jobberman Ghana	1	1	13339246725000000	
12	https://www.jobberman.com.gh/jobs?q=accountant	Accountant Jobs in Ghana Jobberman	1	0	13339246950000000	
13	https://www.google.com/search?q=how+to+export+quickbooks+company+data	how to export quickbooks company data - Google Search	1	0	13339519840000000	
14	https://www.dropbox.com/upload	Upload files - Dropbox	1	0	13339521200000000	
15	https://www.google.com/search?q=starting+an+accounting+firm+in+ghana	starting an accounting firm in ghana - Google Search	1	0	13339522500000000	
16	https://www.ebay.com/sch/i.html?from=R40&trksid=p4432023.m570.11313&_nkw=laptop...	laptop for sale eBay	1	0	13339990937000000	
17	https://www.ebay.com/itm/254654754368?hash=item3b4a9b2e40	Lenovo ThinkPad P50s Laptop 15.6" Core i7 16GB Ram 512GB SSD Win10 - £589.99 eBay	1	0	13339990943000000	
18	https://accounts.google.com/v3/signin/identifier?continue=https%3A%2F%2Fmail.google...	Gmail	1	1	13339990952000000	
19	https://accounts.google.com/ServiceLogin?service=mail	Gmail	1	0	13339990952000000	
20	https://mail.google.com/mail/u/0/	Inbox - kwabena.asare.gh@gmail.com - Gmail	1	0	13339990952000000	

5.4 Intent versus accident (typed vs. link navigation)

Observation: Of the two job-search sites in the history, jobberman.com.gh was reached by direct address entry (typed_count = 1).

linkedin.com/jobs was reached only via a search-result link.

Evidence: Web History transition data for both URLs.

Corrected UTC timestamps for the eBay rows.

Dropbox upload confirmation screenshot.

5.7 Purchase confirmation

Evidence: Receipt_eBay_ThinkPad_P50s.png, order #12-11834-95027, ordered 19 Sep 2023 09:56 UTC. Item total GBP 634.19 with import charges, GBP 589.99 without, matching the listing viewed in Finding 5.5.



Evidence: (1) Receipt_RGD_Business_Registration.png and RGD_Business_Name_Application.pdf, business name registration filed 18 September 2023,

corroborated by the “starting an accounting firm in ghana” search the same day. (2) Sankofa_Accounting_Business_Plan.pdf, a business plan for the registered entity, SANKOFA ACCOUNTING & ADVISORY. (3) Resignation_Letter_DRAFT.pdf, drafted 20 September 2023, two days after the registration filing, not yet sent, with an effective resignation date of 6 October 2023.

The screenshot shows the Autopsy 4.22.1 interface. On the left, the 'Data Sources' pane shows a tree structure of files. The main pane displays a table of files with columns: Name, S, C, O, Modified Time, Change Time, Access Time, Created Time, Size, Flags(Di), Flags(Meta), Known, and Location. The table lists several files, including 'Sankofa_Accounting_Business_Plan.pdf' and 'Resignation_Letter_DRAFT.pdf'. Below the table, the 'Preview' pane shows a document titled 'Registrar General's Department'. The document is a business name registration receipt for 'SANKOFA ACCOUNTING & ADVISORY'. It includes fields for 'Proposed business name', 'Applicant', 'Applicant staff email', 'Nature of business', 'Principal place of business', 'Date of registration', 'Application ref', and 'Fee paid'. The receipt is signed by 'Kwabena Asare' on 18 September 2023.

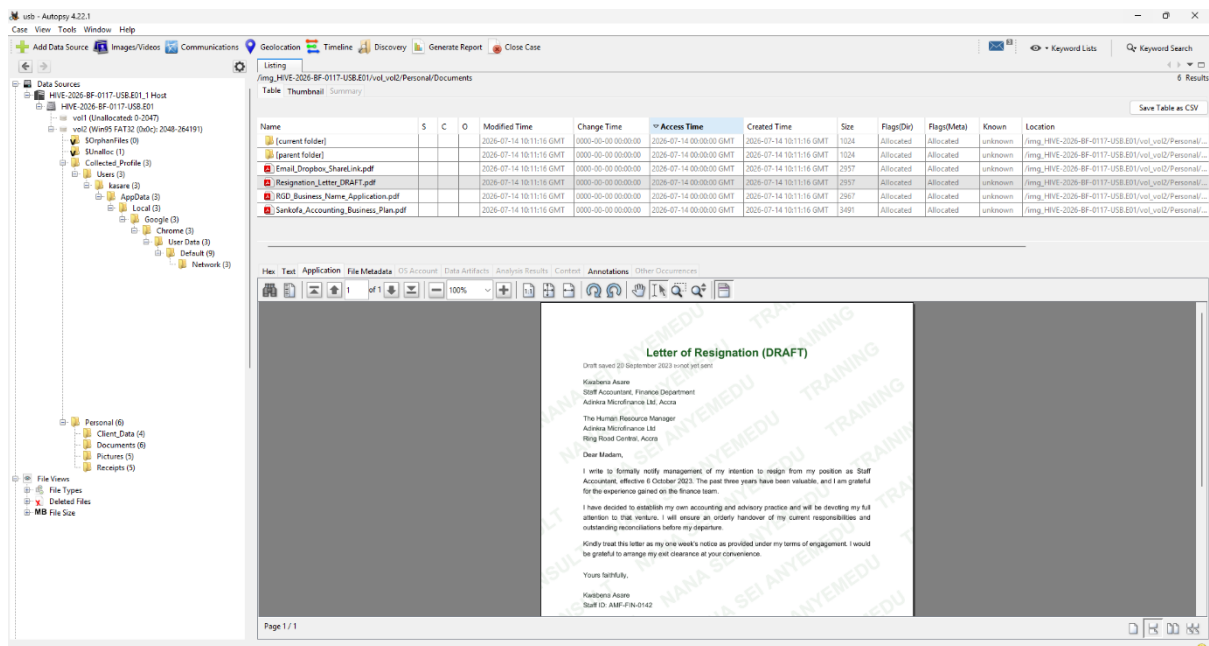
Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Di)	Flags(Meta)	Known	Location
[current folder]				2026-07-14 10:11:16 GMT	0000-00-00 00:00:00	2026-07-14 10:11:16 GMT	2026-07-14 10:11:16 GMT	1024	Allocated	Allocated	unknown	/img_HIVE-2026-BF-0117-USB.E01/vol_vo0/Personal/...
[parent folder]				2026-07-14 10:11:16 GMT	0000-00-00 00:00:00	2026-07-14 10:11:16 GMT	2026-07-14 10:11:16 GMT	1024	Allocated	Allocated	unknown	/img_HIVE-2026-BF-0117-USB.E01/vol_vo0/Personal/...
Email_Desktop_ShareLink.pdf				2026-07-14 10:11:16 GMT	0000-00-00 00:00:00	2026-07-14 10:11:16 GMT	2026-07-14 10:11:16 GMT	2957	Allocated	Allocated	unknown	/img_HIVE-2026-BF-0117-USB.E01/vol_vo0/Personal/...
Resignation_Letter_DRAFT.pdf				2026-07-14 10:11:16 GMT	0000-00-00 00:00:00	2026-07-14 10:11:16 GMT	2026-07-14 10:11:16 GMT	2957	Allocated	Allocated	unknown	/img_HIVE-2026-BF-0117-USB.E01/vol_vo0/Personal/...
RGD_Business_Name_Application.pdf				2026-07-14 10:11:16 GMT	0000-00-00 00:00:00	2026-07-14 10:11:16 GMT	2026-07-14 10:11:16 GMT	2967	Allocated	Allocated	unknown	/img_HIVE-2026-BF-0117-USB.E01/vol_vo0/Personal/...
Sankofa_Accounting_Business_Plan.pdf				2026-07-14 10:11:16 GMT	0000-00-00 00:00:00	2026-07-14 10:11:16 GMT	2026-07-14 10:11:16 GMT	3491	Allocated	Allocated	unknown	/img_HIVE-2026-BF-0117-USB.E01/vol_vo0/Personal/...

Business name registration receipt.

The screenshot shows the Autopsy 4.22.1 interface. On the left, the 'Data Sources' pane shows a tree structure of files. The main pane displays a table of files with columns: Name, S, C, O, Modified Time, Change Time, Access Time, Created Time, Size, Flags(Di), Flags(Meta), Known, and Location. The table lists several files, including 'Sankofa_Accounting_Business_Plan.pdf' and 'Resignation_Letter_DRAFT.pdf'. Below the table, the 'Preview' pane shows a document titled 'SANKOFA ACCOUNTING & ADVISORY'. The document is a business plan for the registered entity, SANKOFA ACCOUNTING & ADVISORY. It includes sections for '1. Concept', '2. Target clients', '3. Start-up requirements', and '4. Timeline'. The 'Start-up requirements' section includes a table with columns 'Item' and 'Estimated cost'.

Item	Estimated cost
Workstation (Lenovo ThinkPad P14s imported)	GHS 7,500
Cloud storage (Google Plus, annual)	GHS 1,450
Business name registration (RGD)	GHS 100
Accounting software licenses	GHS 2,300
Working capital (3 months)	GHS 9,000

Business plan document.



Resignation letter draft.

Interpretation: These are concrete, dated actions rather than idle browsing, and place the registration, the exfiltration, and the resignation draft within a single five-day window.

5.9 Exfiltration channel: self-addressed email

Observation: A self-addressed email shows how the exfiltrated data left the organisation and ties directly to the Dropbox activity in the browser history.

Evidence: Email_Dropbox_ShareLink.pdf, sent from and to kwabena.asare.gh@gmail.com on 18 September 2023 at 14:41:07 UTC, subject “backup”, sharing a Dropbox link to Client_Master_List_Adinkra.xlsx and noting that AdinkraMFI_Q3_Backup.qbb was also uploaded to the same folder. The email states: “Keeping a copy off the work machine before I hand in notice.”

Interpretation: The email, the Dropbox upload screenshot (Finding 5.6), and the Web History entry for dropbox.com/upload are three independent artifacts describing the same event, sent seven minutes after the upload page was visited, which is strong corroboration that the upload was deliberate and that the subject intended to resign shortly afterward.

5.10 Timestamp discrepancy resolved (tool limitation, not evidence tampering)

Observation: An apparent date/time inconsistency identified during initial review (affecting the Dropbox first-access date and the eBay session in Findings 5.3 and 5.5) was investigated and resolved.

Evidence: Chrome/WebKit stores browser timestamps as microseconds since 1601-01-01, not the Unix epoch. Autopsy's raw SQLite table viewer (File Metadata > Application > Table: urls) does not decode this correctly and renders a garbled six-digit year. Re-extracting the History database directly and decoding it with the query datetime(last_visit_time/1000000-

The screenshot shows a Kali Linux virtual machine environment. The top bar indicates the system is running on VMware Workstation. The main application window is Wireshark, which has captured network traffic on the interface 'vnet {vmnic FAT32 (NIC)} 2048-264191'. The packet list contains several entries, with the selected packet being a DNS request from 192.168.1.101 to 192.168.1.1. The packet details pane shows the following information:

- Ethernet II**: Src=VMXNET3 NIC #0, Dst=VirtualBox__NAT__NATAdapter1__en{}
- Internet Protocol Version 4**: Src=192.168.1.101, Dst=192.168.1.1
- Hypertext Transfer Protocol**: GET / HTTP/1.1
- DNS**: Standard query type 1 (query) for google.com

The packet bytes pane shows the raw hex and ASCII data for the selected packet.

Interpretation: The discrepancy originated in the examination tool's raw table view, not in the underlying evidence. No time-stomping or anti-forensic activity is indicated. Browser activity and receipts are tightly self-consistent throughout the case.

Most of the evidence in this case was straightforward. A handful of items were not, and each one is documented here rather than dismissed. The principle applied throughout this examination is that an unanswered question about the evidence is not automatically a sign of tampering, but left unexamined, it becomes a legitimate line of cross-examination in any adversarial proceeding. Every item below was investigated to a documented conclusion, whether that conclusion is a resolution or an acknowledged open question.

Observation: the file sizes shown in the Dropbox upload screenshot do not match the sizes of the corresponding files recovered from the USB drive: Client_Master_List_Adinkra.xlsx is shown as 471 KB in the screenshot versus 7,057 bytes (6.9 KB) recovered, AdinkraMFI_Q3_Backup.qbb is shown as 18.0 MB versus 3,145,853 bytes (3.0 MB) recovered.

Observation: the file recovered from the USB drive is named Adinkra_Client_Master_List.xlsx. The Dropbox upload screenshot and the self-addressed email both reference Client_Master_List_Adinkra.xlsx, the same two words in reversed order. Combined with the size mismatch in 6.1, this warranted further verification.

Verification attempted: hash comparison, the most reliable way to confirm whether two files are identical regardless of naming, could not be performed here, because only the USB-resident file has recoverable byte content. The Dropbox screenshot is an image of a UI, not a file, so there is nothing on that side to hash. The recovered file's hashes are recorded for the file that does exist. Adinkra_Client_Master_List.xlsx, MD5 40f8e5106e0f5ee96b00015070303552, SHA256 22f76a716b123aad7bf39a857456ef58ad0fa7497d39361e1df83837e1771f47.

A naming difference alone does not change a file's hash, hashes depend only on byte content, so the reversed word order by itself proves nothing. However, the size mismatch in 6.1 means these two references cannot be byte-identical regardless of name.

6.3 FAT32 vs. NTFS expectations

Observation: the USB drive is formatted FAT32 (confirmed in Section 2), meaning artifacts such as \$MFT, \$LOGFILE, \$USNJRNL, and Windows registry hives do not exist on this volume.

This is an expected property of the filesystem, not evidence of tampering or an examination gap. Understanding the underlying filesystem before analysis prevents an investigator from chasing artifacts that were never going to be there.

6.4 Evidence labels require context

Observation: the eBay laptop delivery photo (Finding 5.7) is watermarked “FIN-WS-07 - evidence photo”, the same identifier as the workstation under investigation. Read literally, this appears to be a timeline contradiction, how could the workstation's identifier appear on a personal item purchased after the fact?

Verification: every other document and image in the exhibit set carries a similar stamp identifying the case, not just this one photo.

The label is most likely an evidence-tracking identifier applied consistently across the synthetic training exhibits, not a literal claim that the ThinkPad and the workstation are the same machine. It is documented here rather than silently corrected, since the underlying convention has not been confirmed directly with the exhibit's originator.

6.5 Verify your tools, not just your evidence

Observation: Autopsy's raw SQLite table viewer displayed Chrome's last_visit_time as an unusable six-digit year (see Finding 5.10 for the full technical explanation).

Verification: rather than treating the displayed date as authoritative, the raw History database was extracted and decoded independently using the WebKit epoch formula. A peer independently loaded the same evidence in a separate session and observed the identical rendering issue, supporting that this is a reproducible tool limitation rather than a one-off configuration problem.

This was a display bug in the examination tool, not a defect in the evidence. The exhibit's creator was also contacted and corroborated the observation.

Examiner's note: *good forensic work is not just about finding evidence, it is about validating every observation, testing the assumptions behind it, and making sure each conclusion can*

withstand scrutiny. That is what keeps an unresolved anomaly from becoming an opening for the defense.

7. Timeline of Events

The timeline below interleaves corrected Web History timestamps with on-disk document and receipt dates. All timestamps are UTC.

Timestamp (UTC)	Event	Source Artifact
2023-09-15 10:05:30	Google search: "forensic accounting jobs accra"	Web History
2023-09-15 10:06:10	linkedin.com/jobs visited (via search-result link)	Web History
2023-09-15 10:18:45	jobberman.com.gh/ visited (typed directly)	Web History
2023-09-15 10:22:30	jobberman.com.gh/jobs?q=accountant visited	Web History
2023-09-18 14:10:40	Google search: "how to export quickbooks company data"	Web History
2023-09-18 14:30:05	dropbox.com/home revisited, logged in	Web History
2023-09-18 14:33:20	dropbox.com/upload visited	Web History
2023-09-18 14:34	Dropbox upload confirmed: Client_Master_List_Adinkra.xlsx and AdinkraMFI_Q3_Backup.qbb uploaded to /Personal/ByKA	Screenshot_Dropbox_Upload_ClientList.png
2023-09-18 14:41:07	Self-addressed email sent, subject "backup", shares the Dropbox link and states "before I hand in notice"	Email_Dropbox_ShareLink.pdf
2023-09-18 14:55:00	Google search: "starting an accounting firm in ghana"	Web History
2023-09-18, not recorded	RGD business name registration filed for SANKOFA ACCOUNTING & ADVISORY, ref RGD/BN/2023/0091273, GHS 100 paid	Receipt_RGD_Business_Registration.png; RGD_Business_Name_Application.pdf
2023-09-19 09:55:37	eBay search: "laptop for sale"	Web History
2023-09-19 09:55:43	eBay item viewed: Lenovo ThinkPad P50s, £589.99	Web History
2023-09-19 09:55:52	Gmail accessed	Web History
2023-09-19 09:55:58	Outlook Live visited	Web History
2023-09-19 09:56	eBay order placed, order #12-11834-95027, total GBP 634.19	Receipt_eBay_ThinkPad_P50s.png

Timestamp (UTC)	Event	Source Artifact
2023-09-19 09:56:24	dropbox.com/ and dropbox.com/home revisited	Web History
2023-09-19 09:57:23–09:58:08	Leisure browsing: Netflix, IMDb, Google, Formula 1. Last page visited overall: “Latest News” at 09:58:08	Web History
2023-09-20, not recorded	Resignation letter drafted, not sent, effective resignation date 6 Oct 2023	Resignation_Letter_DRAFT.pdf

Note: the ThinkPad was delivered 3 October 2023 (IMG_20231003_ThinkPad_delivered.jpg), within the eBay receipt's estimated delivery window of 2 to 9 October 2023. This falls outside the 15 to 20 September window above but is included for completeness, as it closes the purchase-to-delivery chain.

8. Conclusion

The evidence confirms that the subject conducted job-search and business-formation research on company time, registered a competing business, and drafted a resignation letter within the reviewed window. Data exfiltration is confirmed to a personal Dropbox account and to the USB drive itself. The client spreadsheet and QuickBooks backup are recoverable in both locations, and the self-addressed email, upload screenshot, and Web History entry corroborate one another independently. The apparent timestamp discrepancy identified during initial review is resolved and attributable to a display limitation in Autopsy's raw SQLite table viewer, not to evidence tampering. Browser activity, login data, and receipts are consistent with each other throughout to within seconds. Of the five items examined in Section 6, three are resolved (the FAT32 filesystem scope, the FIN-WS-07 evidence label, and the Autopsy timestamp display bug), two remain open and would benefit from clarification with the exhibit's originator. The file-size mismatch and the filename-order inconsistency between the Dropbox-side records and the USB-resident file, for which hash comparison could not be performed because only one side of the comparison has recoverable byte content.

9. Exhibits / Appendices

- **Appendix A:** Hash verification output and the WebKit-epoch SQL query, `datetime(last_visit_time/1000000-11644473600,'unixepoch')`, run against the extracted History database.
- **Appendix B:** Screenshots are embedded inline within Section 5 alongside the finding they support; full-resolution originals are available in the source E01 image.
- **Appendix C:** Not applicable. This case involves no network-based or malware indicators of compromise, behavioral indicators are documented in Section 5.